CM2015 HT25 Project Carrier Course for Medical Engineers, part 1

Extending the dose rate limit of OpenDosimeter

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Introduction

The OpenDosimeter is an open-source dosimeter used for the monitoring of radiation. Currently available dosimeters that are based on thermoluminescence (TLDs) or optically stimulated luminescence (OSL) are cost-effective and scalable and are often offered as a subscription service by an external party. However they do not provide real-time feedback and involve complicated logistical challenges of sending them to remote facilities for reading. Other real-time electronic dosimeters on the market have high prices, making them unsuitable for radiation monitoring in low-resource communities. As an open source project and a radiation monitoring device, there are numerous ethical issues to consider [1].

Ethical Analysis

1. Safety
    a. Risk of harm from inaccurate reading
    Currently, the OpenDosimeter fails to read high doses of over 1mSv/h, a limitation of the photon-counting approach. Inaccurate readings can compromise occupational dose monitoring and lead to unrecognized overexposure. Physical harm or injury may occur due to exposure to ionizing radiation resulting from device malfunction.
2. Data protection, Privacy, and Governance (GDPR)
    a. What counts as personal/sensitive data
    As per GDPR guideline [5], any related data that is linked to an identifiable person is personal. Therefore, all radiation data taken from a dosimeter that will be used in any type of research or in radiation protection must be anonymized and protected.
    b. Legal/Ethical review
    Any research that is involving person and/or personal/sensitive data must require approval from Swedish Ethical Review Authority (Etikprovningsmyndigheten).
    c. Mitigation and control rules
        - Range limitation: This device is not intended for more than 1mSv/h use. When reaching the limit, the device will show an unreliable warning.
        - Calibration: Regular calibration of dosimeters is required according to Article 34 of Council Directive 2013/59/EUROTOM: “Member States shall require undertakings to seek advice from a radiation protection expert within their areas of competence as outlined in Article 82, on the issues below that are relevant to the practice: (...)
            - (e) regular calibration of measuring instruments and regular checking that they are serviceable and correctly used.” [2]
3. Regulatory boundaries and Intended use
    a. Medical device vs Radiation protection tool

- According to Article 2(1) of MDR - Definition of a medical device (Regulation (EU) 2017/745): "'medical device' means any instrument, (...), for human beings for one or more of the following specific medical purposes:
    - diagnosis, prevention, monitoring, prediction, prognosis, treatment or alleviation of disease,
    - diagnosis, monitoring, treatment, alleviation of, or compensation for, an injury or disability, (...)" [3]

  and therefore, a staff dosimeter which does not monitor a disease or patient condition is excluded from MDR and not considered a medical device.
- According to Article 32 of Council Directive 2013/59/EUROTOM: "Member States shall ensure that the operational protection of exposed workers is based, in accordance with the relevant provisions of this Directive, on: (...)
    - (d) control measures and monitoring relating to the different areas and working conditions, including, where necessary, individual monitoring; (...)" [2]

  And therefore, personal (staff) dosimeters are governed under radiation-protection law.

b. Institutional approval
- Institutional approval for radiation monitoring devices refers to the IEC international standards for all electrical, electronic, and related technologies.
- Institutional approval of personal dosimeters:
    - IEC 61526:2024 Active personal dosimeter, IEC 61674:2024 Diagnostic X-ray dosimeter, and IEC 60846-1:2009 Radiation protection instrumentation.

4. Accuracy, transparency, and scientific integrity
    a. Bias and misuse risk
    The device has 25% error in the photon counting mode while the dose is under 1mSv. Since it is not an intensive exposure to X-ray at that dose, it is quite acceptable. Misuse, such as operating outside the valid range, may lead to incorrect outputs, potentially exposing the user to an unexpectedly high dose.
    b. Open source responsibility
    Open source contributors are responsible for the complete documentation and scientific validation of their solutions. Complete and comprehensive steps for the assembly of the device. The results shall be then transparent and publicly available. According to the EU Cyber Resilience Act which will apply from 11 December 2027 [4], open source projects must include a cybersecurity policy, even if the publisher does not intend to sell the open source device or software by themselves. The current version of the device does not have any existing cybersecurity policies or protections, which could be improved in the next version, especially with future plans of bluetooth or IoT connectivity for the device.
5. Environmental responsibility
    a. Open source hardware itself can lower environmental impact, but must be guided through a framework for sustainable assembly, use, and end-of-life processes as outlined in [6]. Therefore, plans for the refurbish, recycling, and disposal of the Li-ion, e-waste, and radioactive material could be included in documentation.

References

1) N. Ger et al., “OpenDosimeter: Open Hardware Personal X-ray Dosimeter,” arXiv (Cornell University), Sep. 2024, doi: https://doi.org/10.48550/arxiv.2409.09993.
2) “EUR-Lex - 32013L0059 - EN - EUR-Lex,” eur-lex.europa.eu. http://data.europa.eu/eli/dir/2013/59/oj (Last accessed: 04.11.2025)
3) “EUR-Lex - 32017R0745 - EN - EUR-Lex,” eur-lex.europa.eu. https://eur-lex.europa.eu/eli/reg/2017/745/oj (accessed: 04.11.2025)
4) “EOLE 2024 – Navigating the New EU Regulatory Landscape in Open Source – EOLE,” Eolevent.eu, Jan. 08, 2025. https://eolevent.eu/eole-2024-navigating-the-new-eu-regulatory-landscape-in-open-source/ (accessed Nov. 04, 2025)
5) European Union, “General Data Protection Regulation,” Apr. 2016. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32016R0679
6) M. Stange and S. Ihlenfeldt, “Sustainability Through Open-Source Hardware: A Review on Sustainability Aspects,” *Procedia CIRP*, vol. 136, pp. 401–406, Aug. 2025, doi: https://doi.org/10.1016/j.procir.2025.08.070.